

Homework 8

Limits: One-Sided, Infinite, and at Infinity

Name: _____

Instructions

Use a **pencil** and **show your reasoning**. Read the vocabulary section first — the exact words matter a lot here. For the “explain” questions, write full sentences. You may use AI as a **teacher**, but try each problem yourself first.

Reading — Limit Vocabulary

One-sided limits. $\lim_{x \rightarrow a^-} f(x)$ is the value $f(x)$ heads toward as x approaches a from the **left** (from below); $\lim_{x \rightarrow a^+} f(x)$ is the same from the **right** (from above).

When the two-sided limit exists. $\lim_{x \rightarrow a} f(x) = L$ only when *both* one-sided limits exist and *agree*. If the two sides disagree, the two-sided limit **does not exist (DNE)**.

“Does not exist” vs. “undefined” — do not mix these up.

- **DNE** is about the *limit*: there is no single value $f(x)$ heads toward (the two sides disagree, or f blows up).
- **Undefined** is about the *function value* $f(a)$: the rule gives no output there (a hole, or division by zero).

These are **independent**. A limit can exist where f is undefined (a hole: the graph heads to L , but the point (a, L) is missing). And f can be defined where the limit DNE (a jump with a filled dot). Always ask the two questions separately.

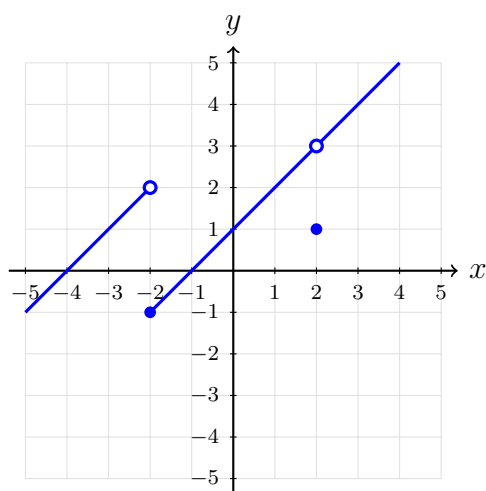
Infinite limits. If $f(x)$ grows without bound near a , we write $\lim_{x \rightarrow a^+} f(x) = +\infty$ (or $-\infty$). The *finite* limit does not exist, but the symbol “ ∞ ” describes *how* it fails — the graph shoots up (or down) along a **vertical asymptote**.

Limits at infinity. $\lim_{x \rightarrow +\infty} f(x)$ and $\lim_{x \rightarrow -\infty} f(x)$ describe the **end behavior** — what height the graph levels off to, far to the right or left. Such a height is a **horizontal asymptote**.

Part 1 — Reading limits from graphs

For each graph, use open circles ($\circ$) as “missing” points and filled circles ($\bullet$) as actual values.

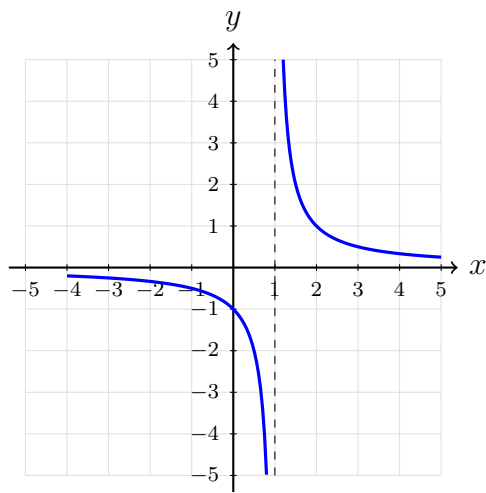
1. Use the graph of f below.



(a) $\lim_{x \rightarrow -2^-} f(x) = \underline{\hspace{2cm}}$ (b) $\lim_{x \rightarrow -2^+} f(x) = \underline{\hspace{2cm}}$ (c) $\lim_{x \rightarrow -2} f(x) = \underline{\hspace{2cm}}$

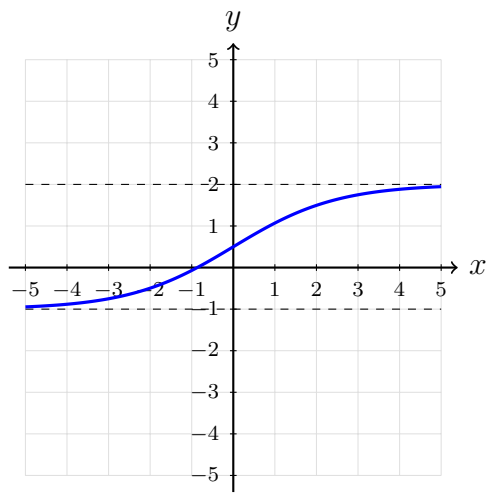
(d) $f(-2) = \underline{\hspace{2cm}}$ (e) $\lim_{x \rightarrow 2} f(x) = \underline{\hspace{2cm}}$ (f) $f(2) = \underline{\hspace{2cm}}$

2. Use the graph of g below (the dashed line is a vertical asymptote).



- (a) $\lim_{x \rightarrow 1^-} g(x) = \underline{\hspace{2cm}}$ (b) $\lim_{x \rightarrow 1^+} g(x) = \underline{\hspace{2cm}}$ (c) $\lim_{x \rightarrow 1} g(x) = \underline{\hspace{2cm}}$
- (d) Is $g(1)$ defined? $\underline{\hspace{2cm}}$ (e) $\lim_{x \rightarrow +\infty} g(x) = \underline{\hspace{2cm}}$

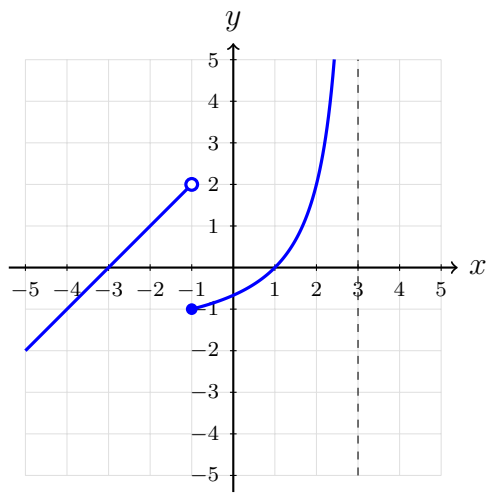
3. Use the graph of h below (the dashed lines are horizontal asymptotes).



(a) $\lim_{x \rightarrow +\infty} h(x) = \underline{\hspace{2cm}}$ (b) $\lim_{x \rightarrow -\infty} h(x) = \underline{\hspace{2cm}}$

- (c) As $x \rightarrow +\infty$, does $h(x)$ ever actually *equal* its limit, or only approach it? Explain.

4. **Putting it together.** Use the graph of f below (the dashed line is a vertical asymptote).



(a) $\lim_{x \rightarrow -1^-} f(x) = \underline{\hspace{2cm}}$ (b) $\lim_{x \rightarrow -1^+} f(x) = \underline{\hspace{2cm}}$ (c) $\lim_{x \rightarrow -1} f(x) = \underline{\hspace{2cm}}$

(d) $f(-1) = \underline{\hspace{2cm}}$ (e) $\lim_{x \rightarrow 3^-} f(x) = \underline{\hspace{2cm}}$

(f) Does $\lim_{x \rightarrow 3} f(x)$ exist as a (finite) number? Explain.

Part 2 — Explain in words

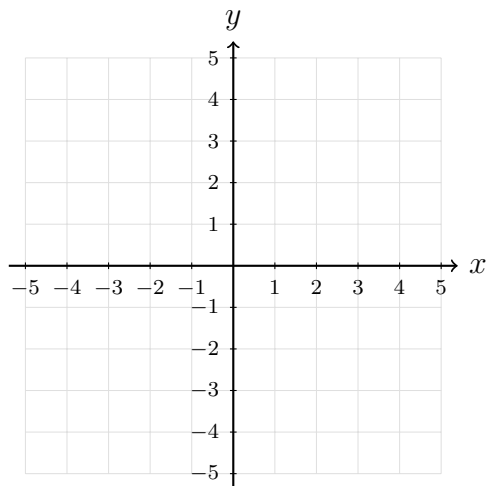
5. Can a limit *exist* at a point where f is *undefined*? Explain, and give (or sketch) an example.

6. Explain the difference between saying “ $\lim_{x \rightarrow a} f(x)$ does not exist” and saying “ $f(a)$ is undefined.” Give an example of each.

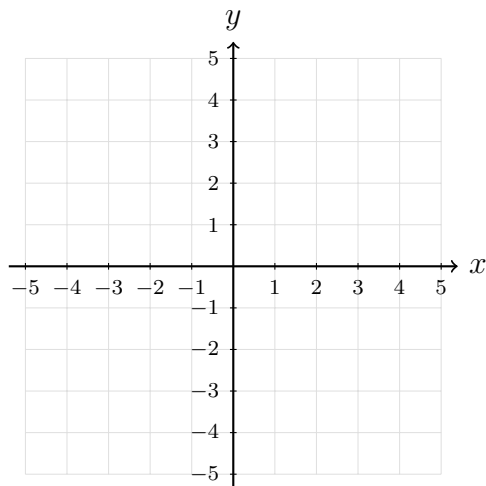
Part 3 — Sketch a graph to match

With a **pencil**, draw **any** function whose graph satisfies all the given conditions. (There are infinitely many correct answers.)

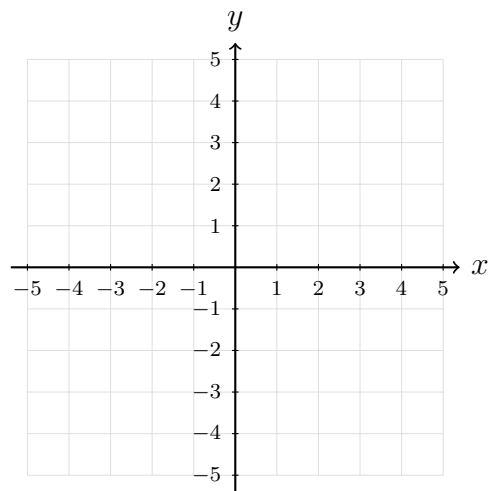
7. $\lim_{x \rightarrow 1^-} f(x) = 3, \quad \lim_{x \rightarrow 1^+} f(x) = -1, \quad f(1) = 0.$



8. $\lim_{x \rightarrow 2} g(x) = 4$ (the two-sided limit exists), but $g(2) = 1.$



9. $\lim_{x \rightarrow 0^-} h(x) = -\infty$, $\lim_{x \rightarrow 0^+} h(x) = +\infty$, and $\lim_{x \rightarrow +\infty} h(x) = 2$.



Part 4 — Find the limit with algebra

First try substituting. If you get an ordinary number, that is the limit. If you get $\frac{0}{0}$, factor and cancel first. For one-sided and infinite limits, think about the sign and size near the point.

10. Find each limit. Show your steps.

(a) $\lim_{x \rightarrow 2} (x^2 + 3x - 1)$

(b) $\lim_{x \rightarrow -3} \frac{x^2 + x - 6}{x + 3}$

(c) $\lim_{x \rightarrow 2} \frac{x^2 - x - 2}{x - 2}$

(d) For $f(x) = \begin{cases} x + 3 & \text{if } x < 1 \\ x^2 + 1 & \text{if } x \geq 1 \end{cases}$, find $\lim_{x \rightarrow 1^-} f(x)$, $\lim_{x \rightarrow 1^+} f(x)$, and $\lim_{x \rightarrow 1} f(x)$.

(e) $\lim_{x \rightarrow 3^+} \frac{1}{x-3}$ and $\lim_{x \rightarrow 3^-} \frac{1}{x-3}$. Does $\lim_{x \rightarrow 3} \frac{1}{x-3}$ exist?

(f) $\lim_{x \rightarrow +\infty} \frac{3x^2 + 2x}{x^2 - 5}$

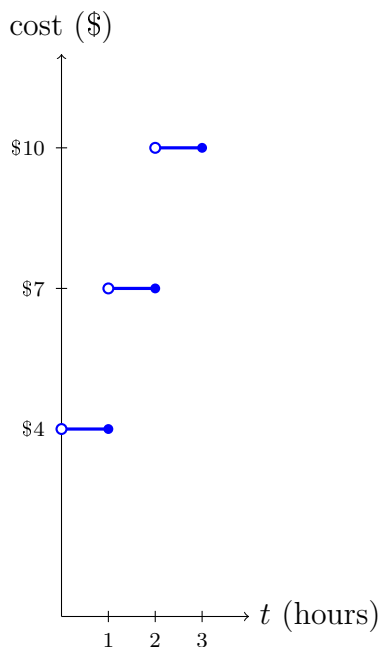
(g) ★ $\lim_{x \rightarrow 0} \frac{\sqrt{x+9} - 3}{x}$ (hint: multiply top and bottom by $\sqrt{x+9} + 3$).

Part 5 — A real-world one-sided limit

A parking garage charges by the hour, and you pay for the whole hour you are in:

$$C(t) = \begin{cases} \$4 & \text{if } 0 < t \leq 1 \\ \$7 & \text{if } 1 < t \leq 2 \\ \$10 & \text{if } 2 < t \leq 3 \end{cases}$$

where t is the number of hours parked. Its graph is a “staircase”:



11. (a) $\lim_{t \rightarrow 2^-} C(t) =$ _____ (b) $\lim_{t \rightarrow 2^+} C(t) =$ _____ (c) $C(2) =$ _____

(d) Does $\lim_{t \rightarrow 2} C(t)$ exist? Explain what the two one-sided limits *mean* for someone parking right around 2 hours.

Notebook Artifact

N2. If you used the internet or AI, describe how you used it.
